Customer Churn Prediction

Model Evaluation and Performance Report

1. Introduction

Customer churn prediction aims to identify customers who are likely to stop using a telecommunications service. In this project, machine learning classification models were trained and evaluated using customer-related information. The main objective was to compare different approaches and determine which model provides the most useful balance between correctly identifying churners and avoiding unnecessary warnings.

2. Dataset and Preprocessing

The dataset contains 7,043 customer records and 21 columns. The target variable is Churn, which was converted into a binary value for classification. The customerID column was removed because it is an identifier rather than a behavioural feature. The TotalCharges column was converted to numeric format, and missing values were handled using the median. The data was divided into training and testing sets using an 80/20 split, stratified by the target variable with random_state=42.

A preprocessing pipeline was used to apply StandardScaler to numerical features and OneHotEncoder to categorical features. Keeping preprocessing inside the pipeline helps ensure that transformations are fitted on training data and then applied consistently to the test data.

3. Model Comparison

Five classification models were evaluated: Logistic Regression, Decision Tree, Random Forest, AdaBoost, and K-Nearest Neighbors (KNN). The comparison considered test accuracy and the precision, recall, and F1-score of the churn class. F1-score was included because accuracy alone can hide weaknesses in identifying the smaller churn class.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.806	0.657	0.559	0.604
Decision Tree	0.729	0.490	0.505	0.497
Random Forest	0.778	0.603	0.476	0.532
AdaBoost	0.802	0.652	0.545	0.594
K-Nearest Neighbors	0.764	0.553	0.575	0.564

Logistic Regression achieved the highest test F1-score (0.604) among the five baseline models, followed by AdaBoost (0.594). KNN achieved a lower F1-score of 0.564, while Decision Tree and Random Forest showed weaker churn detection on the test set. The results suggest that the most complex model is not necessarily the most effective on unseen data.

4. Hyperparameter Tuning

GridSearchCV was used with five-fold cross-validation to search for better parameter settings. The search used F1-score as its scoring metric so that model selection considered performance on the churn class rather than accuracy alone.

4.1 K-Nearest Neighbors

The KNN search explored different values of n_neighbors, distance metrics, and weighting strategies. The selected configuration was n_neighbors=15, metric='manhattan', and weights='uniform'. The tuned model achieved approximately 0.77 test accuracy and an F1-score of 0.56. This is close to the baseline KNN result, so the tuning process did not produce a clear improvement in churn detection on this test split.

4.2 Gradient Boosting

The Gradient Boosting search considered the number of estimators, learning rate, and maximum tree depth. The best configuration recorded in the notebook was n_estimators=200, learning_rate=0.1, and max_depth=2. Its test accuracy was approximately 0.80 and its churn F1-score was 0.58. This is a reasonable result, but it remains below the Logistic Regression baseline F1-score of 0.604.

5. Interpretation of Results

The results highlight why several evaluation metrics are needed. Accuracy measures the proportion of all predictions that are correct, while precision indicates how many customers predicted to churn actually churned. Recall measures the proportion of actual churners identified by the model. F1-score combines precision and recall into a single metric.

For Logistic Regression, precision of 0.657 means that about 65.7% of customers flagged as likely to churn were actual churners. Recall of 0.559 means that the model identified about 55.9% of all customers who churned. Therefore, the model still misses a considerable portion of churn cases, even though its overall accuracy is above 80%.

The Decision Tree and Random Forest results also indicate a possible overfitting issue in the configurations tested: their training performance was much stronger than their test performance. This suggests that model complexity and generalization should be examined during further tuning.

6. Recommendations

The current results support using Logistic Regression as the reference model for further experiments because it achieved the strongest test F1-score among the baseline models. Before selecting a final model, the following steps are recommended:

• Review false-negative cases to understand which churners the model fails to identify.

• Compare models using cross-validation and keep the test set reserved for final evaluation.

• Test different classification thresholds if missing a churner is more costly than contacting a customer who would stay.

• Evaluate the cost and success rate of retention campaigns before using predictions in a real business process.

7. Conclusion

Among the baseline models evaluated, Logistic Regression achieved the best test F1-score, while AdaBoost produced a similar result. Hyperparameter tuning improved the Gradient Boosting configuration to a churn F1-score of approximately 0.58, but it did not exceed the baseline Logistic Regression result. The next stage of the project should focus on improving churn recall, validating performance across different splits, and assessing the business value of model predictions.

Note: The reported metrics reflect the results recorded in the project notebook. Further validation would be required before using the model for operational decisions.